

Supplement to “Spatial aggregation confounds null model-based inference of species association networks”

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S1 Reproducibility

The results of this study can be fully reproduced by installing the package *SpatAggNet* written in R (R Core Team 2018) and available on GitHub. *SpatAggNet* has a dependency on the custom package *pika* also available from GitHub. *pika* must be installed first for *SpatAggNet* to install.

```
pika_pack <- require(pika)
if(!pika_pack) install_github("ajrominger/pika")

this_pack <- require(SpatAggNet)
if(!this_pack) install_github("ajrominger/SpatAggNet")
```

The *SpatAggNet* package includes documented (Wickham *et al.* 2018) and unit-tested (Wickham 2011) functions to carry out all analyses.

The following R computing environment is used for all simulations and analyses:

```
library(SpatAggNet)
library(pika)
library(dplyr)
library(tidyr)
library(ggplot2)
library(cowplot)
library(knitr)
library(parallel)
```

```
# threading defaults
nthrd <- detectCores()
nthrd <- ifelse(round(nthrd * 0.8) >= nthrd - 1 & nthrd > 1,
               nthrd - 1,
               round(nthrd * 0.8))

sessionInfo() |>
  print(locale = FALSE)
```

R version 4.5.2 (2025-10-31)
Platform: aarch64-apple-darwin20
Running under: macOS Sequoia 15.0.1

Matrix products: default
BLAS: /System/Library/Frameworks/Accelerate.framework/Versions/A/Frameworks/vecLib.framework/Versions/A/Libraries/libBLAS.dylib
LAPACK: /Library/Frameworks/R.framework/Versions/4.5-arm64/Resources/lib/libRlapack.dylib;

attached base packages:
[1] parallel stats graphics grDevices utils datasets methods
[8] base

other attached packages:
[1] knitr_1.50 cowplot_1.2.0.9000 ggplot2_4.0.1 tidyr_1.3.1
[5] dplyr_1.1.4 SpatAggNet_0.1.0 pika_1.0

loaded via a namespace (and not attached):
[1] vctrs_0.6.5 cli_3.6.5 rlang_1.1.6 xfun_0.52
[5] purrr_1.1.0 generics_0.1.4 S7_0.2.1 jsonlite_2.0.0
[9] glue_1.8.0 htmltools_0.5.8.1 scales_1.4.0 rmarkdown_2.29
[13] grid_4.5.2 evaluate_1.0.4 tibble_3.3.0 fastmap_1.2.0
[17] yaml_2.3.10 lifecycle_1.0.4 compiler_4.5.2 RColorBrewer_1.1-3
[21] pkgconfig_2.0.3 rstudioapi_0.17.1 farver_2.1.2 digest_0.6.39
[25] R6_2.6.1 tidyselect_1.2.1 pillar_1.11.0 magrittr_2.0.4
[29] withr_3.0.2 gtable_0.3.6 tools_4.5.2

The quarto (Allaire *et al.* 2025) document used to generate this supplement can be found with the R command `system.file("ms/SpatAggNet_supp.qmd", package = "SpatAggNet")`.

S2 Computing spatial abundance distributions

The data included with the *SpatAggNet* package are from Calatayud *et al.* (2019) and take the form of a list of site by species matrices.

```
# load list of site by species matrices provided in SpatAggNet
data("abund_mats")

# remove decimals and empty rows/columns
obs_dat <- lapply(abund_mats, function(x) {
  x <- ceiling(x)
  x <- x[rowSums(x) > 0, colSums(x) > 0]

  return(x)
})
```

In order to investigate the affect of intra-specific spatial aggregation (ISSA) on inferred networks, we first calculate the parameters of the species abundance distribution (SAD) for each data set and the intra-specific spatial abundance distributions (SSAD) for each species in each data set. The `nb_fit` function from *SpatAggNet* computes SSAD parameters as well as summary statistics including goodness of fit.

```
# SSAD statistics for each species in each data set
ssad_stats <- mclapply(obs_dat,
  mc.cores = nthrd,
  FUN = function(x) {
    nb_info <- nb_fit(x)
    cbind(nsite = nrow(x), nspp = ncol(x),
      J = sum(x), nb_info)
  }) |>
  bind_rows(.id = "study")
```

S2.1 Prevalence of the negative binomial SSAD

We would like to know how prevalent the negative binomial SSAD is in the data studied here. This is complicated by the fact that the negative binomial likelihood function does not have a finite maximum when the mean of the data is greater than or equal to the variance. In the data studied here, we see the greatest proportion of cases where $\mathbb{E}[x] \geq \text{Var}[x]$ for rare species and when the number of spatial replicates is small Supp. Figure S1. Unstable estimation for rare species with small sample size is to be expected, but combined with the fact that most species are rare, it is a challenge to assess the prevalence of the negative binomial SSAD.

```

abund_nsite <- expand.grid(abund = 1:40, nsite = 10:90)

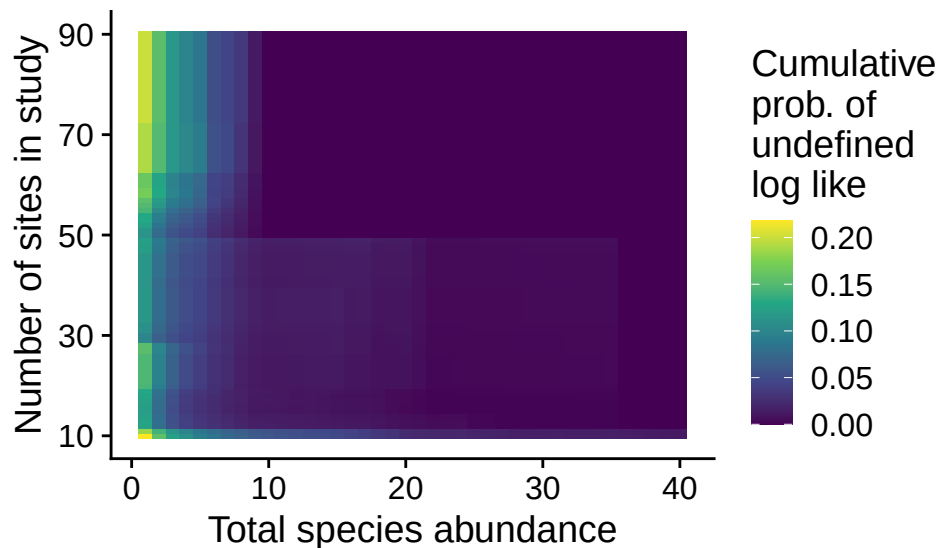
bad_k <- sapply(1:nrow(abund_nsite), function(i) {
  j <- ssad_stats$abund > abund_nsite$abund[i] &
    ssad_stats$nsite > abund_nsite$nsite[i]

  mean(!is.finite(ssad_stats$size[j]))
})

prop_finite_k <- data.frame(abund_nsite, bad_k)

ggplot(prop_finite_k, aes(abund, nsite, fill = bad_k)) +
  geom_tile() +
  scale_fill_viridis_c(name = "Cumulative\nprob. of\nundefined\nlog like") +
  theme_cowplot() +
  scale_y_continuous(breaks = c(10, 30, 50, 70, 90)) +
  xlab("Total species abundance") +
  ylab("Number of sites in study")

```



Supp. Figure S1: Cumulative probability of encountering a species with an undefined log likelihood function. Probability accumulates from right to left and top to bottom. This highlights the difficulty of fitting a negative binomial distribution to rare species in data sets with few sites.

I therefore take two approaches to assessing the prevalence of the negative binomial SSAD:

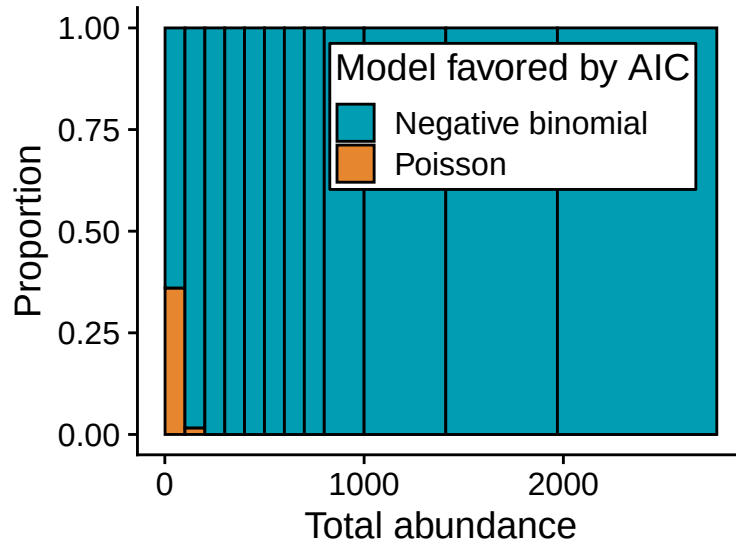
1. where the likelihood is defined, I compare the support for a negative binomial versus a Poisson model with AIC
2. for all the data, whether the likelihood is undefined or defined, I evaluate the goodness of fit of the negative binomial model to the data (described in more detail below)

```
mod_select <- filter(ssad_stats,
                     # only cases with defined likelihood
                     is.finite(size)) |>

# bin by abundance
mutate(abund_bin = cut(abund, c(seq(1, 800, length.out = 9),
                           exp(seq(log(1000), log(max(abund)),
                                length.out = 4)))) |>

group_by(abund_bin) |>
# calculations to plot bars with nb on top of pois
summarize(pois_max = mean(delta_AIC >= 0),
          nb_max = mean(delta_AIC < 0) + pois_max,
          pois_min = 0,
          nb_min = pois_max) |>
ungroup() |>
# resolve upper and lower bounds of rects from
# output of `cut`
pivot_longer(cols = -abund_bin,
              names_to = c("model", ".value"),
              names_sep = "_") |>
mutate(abund_max = gsub(".*|\\]", "", abund_bin) |>
       as.numeric(),
       abund_min = gsub("\\(|,\\.+", "", abund_bin) |>
       as.numeric())

ggplot(mod_select, aes(xmin = abund_min, xmax = abund_max,
                      ymin = min, ymax = max, fill = model)) +
  geom_rect(color = "black", linewidth = 0.5) +
  scale_fill_discrete(name = "Model favored by AIC",
                     labels = c("Negative binomial", "Poisson"),
                     palette = hsv(c(0.52, 0.08), c(1, 0.8), c(0.7, 0.9))) +
  xlab("Total abundance") +
  ylab("Proportion") +
  theme_cowplot() +
  theme(legend.position = c(0.92, 0.92),
        legend.justification = c("right", "top"),
        legend.background = element_rect(fill = "white", color = "black"),
        legend.margin = margin(t = 2, r = 2, b = 2, l = 2))
```



Supp. Figure S2: Proportion of species favoring a negative binomial or Poisson distribution. Model selection is shown across bins of total species abundance. Bins are 100 wide from 0 to 800, then increase in width exponentially because fewer species have very large abundances.

Supp. Figure S2 shows that the negative binomial model is preferred over the Poisson an overwhelmingly majority of the time. Only for rare species is the Poisson occasionally favored by AIC. Even in those cases where Poisson is favored the average ΔAIC separating Poisson and negative binomial is 1.92 with a maximum of 2.07, meaning even when the Poisson is favored, it is only marginally favored.

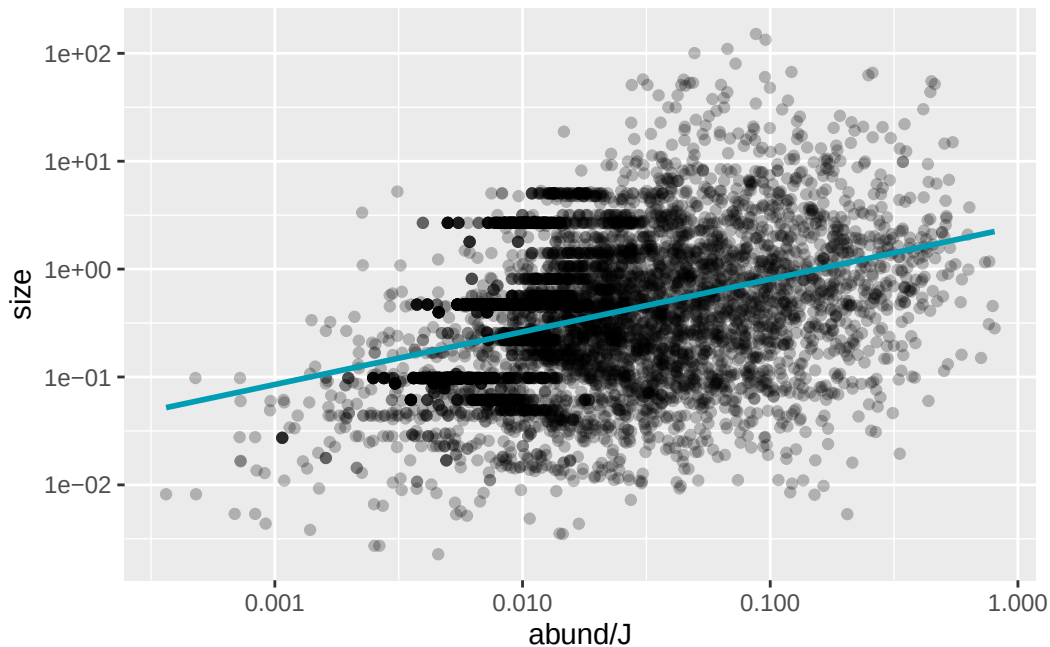
In Supp. Figure S3 I also investigate the relationship of the clustering parameter k with species relative abundance. The purpose of this analysis is to be able to simulate random but realistic data.

```
k_mod <- lm(logk ~ loga,
            data = filter(ssad_stats, is.finite(size)) |>
              mutate(loga = log(abund / J),
                     logk = log(size)))

# a function to randomly draw a k value given abundance
k_fun <- function(abund) {
  J <- sum(abund)

  exp(predict(k_mod, newdata = data.frame(loga = log(abund / J))) +
        rnorm(length(abund), sd = summary(k_mod)$sigma))
}
```

```
ggplot(filter(ssad_stats, is.finite(size)),
  aes(abund / J, size)) +
  geom_point(alpha = 0.25) +
  scale_x_log10() +
  scale_y_log10() +
  geom_smooth(method = "lm", se = FALSE,
    color = hsv(0.52, 1, 0.7))
```



Supp. Figure S3

S3 Computing species abundance distributions

The package *pika* provides functions for fitting common species abundance distribution models. I use the `fitSAD` from *pika* to perform model selection on three standard SAD forms—the log-series, Poisson log-normal, and zero-truncated negative binomial—and record the best fit model and its parameter(s) for each data set.

```
# models to consider
mods <- c('lseries', 'plnorm', 'tnegb')

# loop over studies, fitting SAD models
sad_stats <- mclapply(obs_dat,
```

```

      mc.cores = nthrd,
      FUN = function(x) {
nsite <- nrow(x)
x <- colSums(x)
s <- fitSAD(x, mods)

# which model is best supported
i <- which.min(sapply(s, AIC))

# mle of params of bets model
o <- s[[i]]$MLE

# if lseries is best, only has 1 param
if(i == 1) o <- c(o, NA)

o <- c(i, o, sum(x), length(x), nsite)
names(o) <- NULL

return(o)
}) |>
do.call(rbind, args = _) |>
as.data.frame()

sad_stats <- cbind(study = rownames(sad_stats), sad_stats)
names(sad_stats)[-1] <- c('mod', 'par1', 'par2', 'J', 'nspp', 'nsite')
rownames(sad_stats) <- NULL
sad_stats$mod <- mods[sad_stats$mod]

sad_select <- filter(sad_stats) |>
  mutate(abund_bin = cut(J, c(seq(1, 1400, length.out = 9),
                                exp(seq(log(2000), log(max(J)),
                                          length.out = 4)))))) |>
  group_by(abund_bin) |>
  summarize(lseries_max = mean(mod == "lseries"),
            plnorm_max = mean(mod == "plnorm") + lseries_max,
            tnegb_max = mean(mod == "tnegb") + plnorm_max,
            lseries_min = 0,
            plnorm_min = lseries_max,
            tnegb_min = plnorm_max) |>
  ungroup() |>
  pivot_longer(cols = -abund_bin,
               names_to = c("model", ".value"),

```

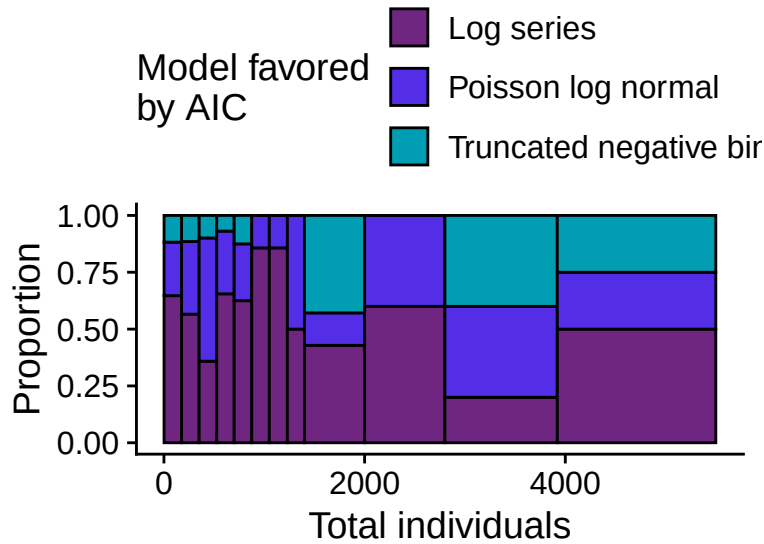
```

      names_sep = "_") |>
mutate(abund_max = gsub(".*|\\]", "", abund_bin) |>
      as.numeric(),
      abund_min = gsub("\\(|,.*", "", abund_bin) |>
      as.numeric())

ggplot(sad_select, aes(xmin = abund_min, xmax = abund_max,
                      ymin = min, ymax = max, fill = model)) +
  geom_rect(color = "black", linewidth = 0.5) +
  scale_fill_discrete(name = "Model favored\nby AIC",
                     labels = c("Log series",
                                "Poisson log normal",
                                "Truncated negative binomial"),
                     palette = hsv(c(0.8, 0.7, 0.52),
                                   c(0.7, 0.8, 1),
                                   c(0.5, 0.9, 0.7))) +

  xlab("Total individuals") +
  ylab("Proportion") +
  theme_cowplot() +
  theme(legend.position = "top") +
  guides(fill = guide_legend(nrow = 3))

```



Supp. Figure S4: Proportion of studies favoring one of three possible species abundance distribution models: log series, Poisson log normal, truncated negative binomial. Model preference is shown across different different total numbers of individuals recorded by each study. Bins start at a width of 175 from 1 to 1,400, then expand in width exponentially. We see there is no consistent pattern for which model is favored across total number of individuals.

S4 Studying networks simulated based on real abundance data

Now we seek to compare the networks reconstructed from real data to those reconstructed from purely random data simulated by sampling SADs and SSADs without any true interaction/association between species.

First we compute network statistics for the real data

```
obs_nets <- mclapply(obs_dat, net_stats, mc.cores = nthrd) |>
  do.call(rbind, args = _) |>
  as.data.frame() |>
  (\(x) {
    x <- cbind(study = rownames(x), x)
    rownames(x) <- NULL
    x
  })()
```

```
ggplot(filter(obs_nets, !is.na(pos_rho) & !is.na(neg_rho)),
  aes(pos_wm_abund - neg_wm_abund)) +
  geom_histogram(bins = 20, alpha = 0.5, color = "black") +
  xlim(-0.33, 0.33)
```

```
ggplot(filter(obs_nets, !is.na(pos_rho) & !is.na(neg_rho)),
  aes(pos_rho - neg_rho)) +
  geom_histogram(bins = 20, alpha = 0.5, color = "black") +
  xlim(c(-2, 2))
```

```
ggplot(filter(obs_nets, !is.na(pos_rho) & !is.na(neg_rho)),
  aes(neg_rho)) +
  geom_histogram(bins = 24, alpha = 0.5, color = "black") +
  xlim(-1, 1)
```

```
sim_nets_dat <- sim_nets(sad_stats,
  mc_cores = nthrd,
  k_fun = k_fun,
  nsim = round(1.25 * sum(!is.na(obs_nets$pos_rho))))
```

References

- Allaire, J.J., Teague, C., Scheidegger, C., Xie, Y., Dervieux, C. & Woodhull, G. (2025). [Quarto](#).
- Calatayud, J., Andivia, E., Escudero, A., Melián, C.J., Bernardo-Madrid, R., Stoffel, M., Aponte, C., Medina, N.G., Molina-Venegas, R., Arnan, X., Rosvall, M., Neuman, M., Noriega, J.A., Alves-Martins, F., Draper, I., Luzuriaga, A., Ballesteros-Cánovas, J.A., Morales-Molino, C., Ferrandis, P., Herrero, A., Pataro, L., Juen, L., Cea, A. & Madrigal-González, J. (2019). Positive associations among rare species and their persistence in ecological assemblages. *Nat Ecol Evol*.
- R Core Team. (2018). *R: A language and environment for statistical computing*. R Foundation for Statistical Computing, Vienna, Austria.
- Wickham, H. (2011). *testthat: Get started with testing*. *The R Journal*, **3**, 5–10.
- Wickham, H., Danenberg, P. & Eugster, M. (2018). *roxygen2: In-line documentation for R*.